

Phishing Detection: NLP and AI

Research Proposal Defense

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Why Phishing Detection Matters

- Phishing remains a leading vector for credential theft, financial fraud, and initial access in larger cyberattacks.
- Attacks increasingly use natural, well-written language and social engineering, evading simple keyword and blacklist filters.
- Generative AI has lowered the barrier to crafting convincing phishing emails and messages at scale.
- NLP and AI offer content-level understanding that static, signature-based defenses lack.

Limitations of existing approaches

- Blacklists and URL heuristics fail against new, unseen domains
- Rule-based filters miss subtle linguistic manipulation
- Shallow lexical ML features are weak to paraphrasing

Gap this work targets

- Limited robustness against AI-generated phishing text
- Need to combine semantic (NLP) and structural (AI/ML) signals
- Few studies benchmark cross-domain generalization (email vs. SMS vs. chat)

Research Objectives

- Design an NLP-based feature representation that captures semantic and stylistic cues of phishing text.
- Develop and compare AI/ML/DL models (classical ML, LSTM, transformer-based) for phishing classification.
- Evaluate robustness against paraphrased and AI-generated phishing samples.
- Benchmark generalization across message types (email, SMS, social/chat).

Proposed Methodology

Stage	Details
Data	Public phishing/legitimate corpora (email, SMS, URL) + synthetic AI-generated phishing samples
Preprocessing	Cleaning, tokenization, URL/header feature extraction
Representation	TF-IDF baseline; contextual embeddings (BERT-family)
Models	Logistic Regression / Random Forest baselines; LSTM; fine-tuned transformer
Evaluation	Accuracy, F1, ROC-AUC; robustness on paraphrased/adversarial subsets

Expected Outcomes

- An NLP+AI phishing detection pipeline more resilient to AI-generated and paraphrased content.
- Comparative benchmark of classical ML vs. deep learning vs. transformer models on phishing text.
- Cross-domain generalization results (email, SMS, chat) to guide practical deployment.
- Publishable methodology and dataset/benchmark contribution for future research.

Timeline & References

Tentative Timeline

- Months 1–2: Literature review & data collection
- Months 3–4: Preprocessing & baseline models
- Months 5–6: Deep learning & transformer models
- Months 7–8: Robustness & cross-domain evaluation
- Months 9–10: Write-up & submission

Key Reference Areas

- Phishing detection surveys (ML/DL-based)
- Transformer-based text classification (BERT, RoBERTa)
- Adversarial robustness in NLP
- Public phishing datasets (e.g., PhishTank-derived corpora)

Thank You

Questions & Discussion